Change of Coordinate Matrix in $\mathbb{R}^n$

Recall that if $\mathcal{B} = \{\mathbf{b}_1, \ldots, \mathbf{b}_n\}$ is a basis of a vector space V, then the matrix

$$P_{\mathcal{B}} = \begin{bmatrix} [\mathbf{b}_1]_{\mathcal{C}} & \ldots & [\mathbf{b}_n]_{\mathcal{C}} \end{bmatrix}$$

Represents our basis vectors as an array of real numbers. Where $\mathcal{C}$ is the "standard" basis for our vector space. (In $\mathbb{R}^n$, these are the columns of the identity matrix)

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$$P_{\mathcal{B}}\mathbf{x} = x_1 [\mathbf{b}_1]_{\mathcal{C}} + \cdots + x_n [\mathbf{b}_n]_{\mathcal{C}}$$

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This means if $\mathbf{x}$ is an array of coordinates for the $\mathcal{B}$ basis, then the output is a vector written in the standard basis!

Change of Coordinate Matrix in $\mathbb{R}^n$ Example

Consider the basis

$$\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} \right\}$$

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Change of Coordinate Matrix in $\mathbb{R}^n$ is a Linear Transformation

Theorem

If $V = \mathbb{R}^n$, then the Coordinate Matrix $P_{\mathcal{B}}$ is a Linear Transformation given by

$$P_{\mathcal{B}} : [\mathbf{x}]_{\mathcal{B}} \mapsto \mathbf{x},$$

which changes a vector from $\mathcal{B}$ coordinates to being written in the standard basis

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$$P_{\mathcal{B}}^{-1} : \mathbf{x} \mapsto [\mathbf{x}]_{\mathcal{B}}$$

changes a vector from the standard basis back to $\mathcal{B}$ coordinates.

Changing Coordinate From One Basis to Another

In some data science applications, we may want to change problems from one basis to another.

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Is there a potentially better way?

A Better Way Without Explicitly Computing an Inverse

Yes!

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We also have that

$$\underset{\mathcal{C}\leftarrow\mathcal{B}}{P} = \begin{bmatrix} [\mathbf{b}_1]_{\mathcal{C}} & \cdots & [\mathbf{b}_n]_{\mathcal{C}} \end{bmatrix}$$

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Which we can solve in one of two ways

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$$\left[\begin{array}{c|c} P_{\mathcal{C}} & P_{\mathcal{B}} \end{array}\right]$$

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Which is solving the equation

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then plugging in $\mathbf{y} = \mathbf{b}_1, \ldots, \mathbf{b}_n$.

Either will give the right answer, it's just a matter of preference!

Change of Coordinate Example

Consider the following bases of $\mathbb{R}^3$.

$$\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} \right\}, \mathcal{C} = \left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$

Compute $\underset{\mathcal{C} \leftarrow \mathcal{B}}{P}$.

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$$\left[\begin{array}{c|c} P_{\mathcal{C}} & P_{\mathcal{B}} \end{array} \right] = \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 2 \\ 1 & 1 & 0 & 2 & 1 & 1 \\ 1 & 1 & 1 & 1 & 2 & 0 \end{array} \right] \to \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 2 \\ 0 & 1 & 0 & 1 & 1 & -1 \\ 0 & 1 & 1 & 0 & 2 & -2 \end{array} \right]$$

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So

$$\underset{\mathcal{C}\leftarrow\mathcal{B}}{P} = \begin{bmatrix} 1 & 0 & 2 \\ 1 & 1 & -1 \\ -1 & 1 & -1 \end{bmatrix}$$

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So the columns of $\underset{\mathcal{C} \leftarrow \mathcal{B}}{P}$ are given by

$$[\mathbf{b}_1]_{\mathcal{C}} = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}, [\mathbf{b}_2]_{\mathcal{C}} = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, [\mathbf{b}_3]_{\mathcal{C}} = \begin{bmatrix} 2 \\ -1 \\ -1 \end{bmatrix}$$

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Compute $\underset{\mathcal{C} \leftarrow \mathcal{B}}{P}$

$$\underset{\mathcal{C} \leftarrow \mathcal{B}}{P} = \begin{bmatrix} 0 & \frac{1}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{1}{4} & \frac{3}{4} \\ \frac{1}{2} & -\frac{1}{4} & \frac{1}{4} \end{bmatrix}$$

Matrix Vector Space Practice

Let V be the vector space of symmetric 2×2 real matrices. We have the basis

$$\mathcal{B} = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}.$$

Consider the set $\mathcal{C} = \left\{ \begin{bmatrix} 1 & -2 \\ -2 & 2 \end{bmatrix}, \begin{bmatrix} 0 & 2 \\ 2 & 4 \end{bmatrix}, \begin{bmatrix} 3 & -1 \\ -1 & 1 \end{bmatrix} \right\}$. Answer the following questions

1) Show that $\mathcal{C}$ is a basis for V.

2) Write the matrix corresponding to $\left[\begin{bmatrix} 2 \\ 3 \\ -2 \end{bmatrix} \right]_{\mathcal{C}}$ in $\mathcal{B}$ coordinates

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in $\mathcal{B}$ coordinates

$$\begin{bmatrix} -4 & 4 \\ 4 & 14 \end{bmatrix}$$